



ALMASOFT
Mer värde i praktiken

Almasoft Tidbok_online REST API

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Alma Tidbok_online REST API



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Content information

Below is a description of the format of the API calls supported by the eTand REST API.

The connection only supports

- HTTP1.1
- TLS1.1 or TLS1.2
- All calls are made using HTTP POST.
- All calls are made using the JSON format.
- HTTP header "Content-Type" must be set to "application/json".

Below is a description of the format of the API calls supported by the eTand REST API.

The character set is

- UTF-8

The format for time entry is

- DateTime: "YYYY-MM-DD HH:MM:SS"
- DateTimeHighRes: "YYYY-MM-DD HH:MM:SS.ZZZ"
- Date: "YYYY-MM-DD"
- Time: "HH:MM:SS"
- TimeHighRes: "HH:MM:SS.ZZZ"

The format for social security numbers is

- Person No: "YYYYMMDDNNNN"

The format for phone numbers is

- Phone number: "099999999999"

Value for customer number

To find out which Almasoft AB customer number a particular clinic has, you can contact the clinic and ask.

- almasoftKund: 0-99999

Value for integrator

To find out what information must be entered for integratorCode and integratorLosen, please contact Almasoft AB.

Server address

The server is the Almasoft AB server that already manages eTand Patientportal today.

- Server address: <https://etand.se>
- Server port: 16443

Test environment

The test/acceptance environment we have is unfortunately only for internal calls. However, we provide a test client/clinic in the production environment that can be used instead.

The address below is to the eTand web portal for patients and can be used to view the results of calls to the eTand REST API. The address below goes to a test clinic with Almasoft customer number 9002.

- Webaddress: <https://9002.etand.se>
- Patient social security number: 19010101-0106
- Patient password: 123456

In the patient portal with the address above, select **Logga in** and then **Mina bokningar** to see bookings made using eTand REST API calls.

It is also possible in the patient portal to see which treatment types and therapists the clinic has. Select Book new (before or after logging in). See the selections for **Behandlingstyper** and after selection of treatment type **Behandlare**.

This customer number 9002 can be used for testing in calls to the eTand REST API and the results of the calls can be seen on the eTand web portal.

Note

- The proposed eTand REST API calls provide the same functionality as in the eTand web portal button Book new (before login).
- The database for test clinic 9002 is reset every night, so no information can be changed/saved for longer than 24 hours.

Calls

[/adm/v1/ping](#)

Called to perform communication control between the caller and the REST service

Request

```
{
  // A code för a certain integrator
  integratorKod      String
  // A password that is valid för a certain integrator
  integratorLosen    String
}
```

Response (OK)

```
{
  // Time indication in the REST server when the call is handled
  pingTid            DateTime
}
```

Response (ERROR)

Only generic error code is used.

Call

[/api/v1/login](#)

Called to log in for authentication and start a session. The response from the call gives an access code that must be entered in other calls.

The returned access code has a lifespan of 1 our.

Request

```
{
  // A number that applies to a certain customer/clinic
  almasoftKund      Integer
  // A code för a certain integrator
  integratorKod      String
  // A password that is valid för a certain integrator
  integratorLosen    String
}
```

Response (OK)

```
{
  // An access code to use in all following calls
  accessKod          String
}
```

```

    // Time indication in the REST server when the call is handled
    loginTid          DateTime
}

```

Response (ERROR)

Felkod	Beskrivning
1	The clinic does not allow external integrators to connect.
2	The clinic does not allow the integrator to connect and log in..

[/api/v1/logout](#)

Called to log out/end the session in the REST server Anropas för att logga ut / avsluta sessionen i REST servern.

Request

```

{
    // En siffra som beskriver en viss Almasoft AB kund/klinik
    almasoftKund      Integer
    // A code för a certain integrator
    integratorKod     String
    // A code received at login
    accessKod         String
}

```

Response (OK)

```

{
    // Time indication in the REST server when the call is handled
    logoutTid        DateTime
}

```

Response (ERROR)

Only generic error code is used.

[/tdl/v1/avbokaTid](#)

Called to cancel (delete) an appointment in the clinics booking system.

Note

- A cause for cancelling must be entered while cancelling the appointment.

Request

```

{

```

```

// A number that applies to a certain customer/clinic
almasoftKund      Integer
// A number that applies to a certain customer/clinic
integratorKod     String
// A code received at login
accessKod         String

// A code for the patient
patientKod        String
// A code for the appointment
tidbokningKod     String
// A text describing why the cancellation is desired
orsak             String
}

```

Response

```

{
  // A code for the appointment (same as in the request)
  tidbokningKod    String
}

```

Response (ERROR)

Felkod	Beskrivning
1	The specified patientkod is incorrect
2	The specified tidbokningskod is incorrect
3	The text describing why a cancellation is made has to be entered. (orsak)
4	An error occurred while cancelling the appointment

[/tdl/v1/behandlare](#)

Called to retrieve the practitioners that are present in the booking system

Request

```
{
  // A number that applies to a certain customer/clinic
  almasoftKund      Integer
  // A code för a certain integrator
  integratorKod     String
  // A code received at login
  accessKod         String

  // A code for treatmenttype
  behandlingstypKod String / ''
}
```

Response (OK)

```
{
  // A list of therapists
  behandlare      Array
  {
    // Therapist code
    kod           String
    // Therapist name
    namn          String
    // Therapist title
    titel         String
  }
}
```

Response (FEL)

Felkod	Beskrivning
1	The specified behandlingstypKod is incorrect

</tdl/v1/behandlingstyper>

Called to retrieve the treatment types that are present in the bookingsystem.

Request

```
{
  // A number that applies to a certain customer/clinic
  almasoftKund      Integer
  // A code för a certain integrator
  integratorKod     String
  // A code received at login
  accessKod         String

  // A code for a therapist
  behandlareKod     String / ''
}
```

Response (OK)

```
{
  // List of found treatment types
  behandlingstyper  Array
  {
    // Treatment type code
    kod              String
    // Treatment type name
    namn             String
    // Treatment type length (in minutes)
    langd            Integer
  }
}
```

Response (FEL)

Felkod	Beskrivning
1	The specified behandlingstyper is incorrect

[/tdl/v1/bokaTid](#)

Called to make an appointment i the booking system.

Note

- Specified value for **tidStart** may only be in the future.
- The call must be preceded by calls to **/tdl/v1/registreraPatient**. This in order to get the code for the patient.

Request

```
{
  // A number that applies to a certain customer/clinic
  almasoftKund      Integer
  // A code för a certain integrator
  integratorKod     String
  // A code received at login
  accessKod         String

  // A code for the patient
  patientKod        String
  // A code for the therapist
  behandlareKod     String
  // A code for the treatment type
  behandlingstypKod String
  // The desired time for the appointment
  tidStart          DateTime
  // Note for the appointment (maxumum size 160 characters)
  anteckning        String / ''
}
```

Response

```
{
  // A code for the appoint
  tidbokningKod     String
}
```

[/tdl/v1/bokaTid \(forts.\)](#)

Response (ERROR)

Felkod	Beskrivning
1	The specified patientKod is incorrect.
2	The specified behandlareKod is incorrect.
3	The specified behandlingstypKod is incorrect.
4	The Specified tidStart is incorrect.
5	The appointment cannot be booked. Selected time may have been taken by someone else

[/tdl/v1/ledigaTider](#)

Called to retrieve the vacant timeslots available within a certain time range.

Note

- Searching for vacant timeslots may take place over a maximum of 1 month.
- The time range must be in the future.

Request

```
{
  // A number that applies to a certain customer/clinic
  almasoftKund      Integer
  // A code för a certain integrator
  integratorKod     String
  // A code received at login
  accessKod         String

  // A code for therapist
  behandlareKod     String / ''
  // A code for treatment type
  behandlingstypKod String
  // The first day of search
  datumStart       Date
  // The last day of search
  datumSlut        Date
}
```

Response (OK)

```
{
  // List with vacant timeslots
  behandlareTider   Array
  {
    //A code for therapist
    behandlareKod   String

    // List with vacant timeslots for a therapist
    ledigaTider     Array
  }
}
```

```

    {
      // Vacant timeslot for a therapist
      tidStart      DateTime
    }
  }
}

```

Response (ERROR)

Felkod	Beskrivning
1	The specified behandlareKod is incorrect.
2	The specified behandlingstypKod is incorrect
3	The specified timer ange is incorrect
4	The clinic demands that behandlareKod is filled out

[/tdl/v1/registreraPatient](#)

Called to register which patient applies. If necessary, the patient is added as a new patient.

Note

- Information must be entered at least in **Email** or **Mobil** (both are allowed).
- If the patient is already in the database, the specified Email or Mobil must match what is already specified in the database.
- If the clinic does not allow child patients to register, error code 3 is returned (the same error code as when the clinic does not allow any patient to register). In that case, the patient needs to contact the clinic to register as a patient.

Request

```

{
  // A number that applies to a certain customer/clinic
  almasoftKund      Integer
  // A code för a certain integrator
  integratorKod     String
  // A code received at login
  accessKod         String

  // The patients social security number
  personnr          String
  // The patients first name
  fnamn             String
  // The patients surname
  enamn             String
  // The patients email address
  email             String / ''
  // The patients mobile number
  mobil             String / ''
}

```

```
}
```

Response

```
{  
  // A code for the patient  
  patientKod          String  
  // A number for the patient in the clinic  
  patientNr           Integer  
  // A status for the patient (B=existing, N=new (in other words, added),  
  // O=unknown)  
  patientStatus       Char  
}
```

[/tdl/v1/registreraPatient](#) (forts.)

Response (ERROR)

Felkod	Beskrivning
1	The patient information provided is incorrect
2	Specified patient Email or Mobile does not match what is in the database
3	The clinic does not allow patients to be managed electronically (for new patients)
4	The clinic does not allow patients to be managed electronically (for existing patients)

Error messages

If an error occurs during API calls, an error response is returned. Most often, the answer is returned as a JSON string, but HTML text can also be given in certain error situations.

Error code is indicated both in the form of HTTP error code and with a JSON error code.

Error message (JSON)

```
{
  // Code for error message
  felKod          Integer
  // Text describing the error message
  felText         String
}
```

Common error codes apply to all calls.

Felkod	Beskrivning
-1	Invalid content of the call
-2	Invalid information specified for any integrator, access code or Almasoft customer
-3	Authorization is missing for the specified integrator and/or Almasoft customer
-4	No response received from counterparty
-5	A general system error has occurred
-6	The function is not implemented

In addition to the above error codes for general errors, specific error indications may occur with each API call. See description above for respective API calls.

Note

There may be new error codes in the future than those documented in this document.

Error message (HTML)

HTTP	Beskrivning	Notering
400	An error occurred	See felKod in the response. See common error codes as well as specific error codes for each API call.
401	Unauthorized	The specified integrator and/or access code lacks authorization. A new login should be attempted.
403	Forbidden	The specified integrator does not have the right to make the call. Retries should not be made from the caller.
404	Cannot find	Calls have been made to an API address that is not implemented.
500	Internal server error	An internal error has occurred in the eTand REST service.
503	Maintenance in progress	The eTand REST service is undergoing maintenance and is temporarily down.

User cases

Below are some use cases described for the eTand REST API.

Perform communication test

This call can be used when you want to check that the eTand REST server is running and responding to calls. For example, it can be a way to perform surveillance.

Call: `/adm/v1/ping`

Retrieve treatment types independent therapist

In this case, you are interested in all treatment types that are available at the clinic, regardless of which therapists perform them.

Call: `/api/v1/login`
 `/tdl/v1/behandlingstyper` (with `behandlareKod = ""`)
 `/api/v1/logout`

Get treatment types for a specific therapist

In this case, you are interested in all types of treatment performed by the specified treatment provider.

Call: `/api/v1/login`
 `/tdl/v1/behandlingstyper` (with `behandlareKod = "<från API behandlare>"`)
 `/api/v1/logout`

Retrieve therapist regardless of treatment type

In this case, you are interested in all therapists available at the clinic, regardless of the types of treatment they perform.

Call: `/api/v1/login`
 `/tdl/v1/behandlare` (with `behandlingstypKod = ""`)
 `/api/v1/logout`

Retrieve therapists for a certain treatment type

In this case, one is interested in the therapists available at the clinic with the type of treatment indicated.

Call: `/api/v1/login`
 `/tdl/v1/behandlare` (with `behandlingstypKod = "<från API behandlingstyper>"`)
 `/api/v1/logout`

Retrieve free time slot independent therapists

In this case, you are interested in free time slots regardless of which therapists are available.

Call: `/api/v1/login`
 `/tdl/v1/ledigaTider` (with `behandlareKod = ""`)
 `/api/v1/logout`

Note

- It may take a little longer to look up free times when all therapists at the clinic have to be searched through! The impact on the clinic's database in the form of increased load will occur and also transport of a larger response over the clinic's internet connection will occur.
- Only searching for free times in the future is allowed.
- Currently, the maximum date range allowed is 1 month.

Retrieve free time slots for a particular therapist

In this case, you are interested in free time slots for a certain therapist.

Call: `/api/v1/login`
 `/tdl/v1/ledigaTider` (with `behandlareKod = "<från API behandlare>"`)
 `/api/v1/logout`

Note

- Only searching for free times in the future is allowed.
- Currently, the maximum date range allowed is 1 month.

Retrieve free time slots with contiguous session

The **accessKod** you get back when calling login for a specified **almasoftKund** has a lifetime of 1 hour today. This means that you can collect a number of eTand REST API calls for the same login.

Call:

```
/api/v1/login  
/tdl/v1/ledigaTider (för t.ex. emergencies)  
/tdl/v1/ledigaTider (för t.ex. revision)  
/tdl/v1/ledigaTider (för t.ex. root canal)  
/api/v1/logout
```

Note

- Only searching for free times in the future is allowed.
- Currently, the maximum date range allowed is 1 month.

Make a reservation

The **accessKod** you get back when calling login for a specified **almasoftKund** has a lifetime of 1 hour today. This means that you can collect a number of eTand REST API calls for the same login.

Call:

```
/api/v1/login  
/tdl/v1/registreraPatient  
/tdl/v1/bokaTid  
/api/v1/logout
```

Note

- When booking an appointment, either the patient's email or the patient's mobile number must be specified (it is also possible that both are filled in).
- If the patient's is not in the database, it is added.
- If the patient is in the database but not in the clinic, it is linked to the clinic.
- If the patient is already in the database for the clinic, nothing happens.
- Several clinics can share the same server and the same database. This is for example to save money or to share a reception.

Book an appointment with a cohesive session

The accessKod you get back when calling login for a specified almasoftKund has a lifetime of 1 hour today. This means that you can collect a number of eTand REST API calls for the same login.

Call:

- `/api/v1/login`
- `/tdl/v1/behandlingstyper`
- `/tdl/v1/behandlare`
- `/tdl/v1/ledigaTider`
- `/tdl/v1/registreraPatient`
- `/tdl/v1/bokaTid`
- `/api/v1/logout`